

Linear Regression

Advanced Statistics Lab

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Linear Regression

Linear Regression deals with fitting predicted values based on input features and regression weights to observations.

In ordinary least squares, we aim to find regression weights that minimize the mean squared error between predicted and observed values:

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (y_i - x_i \beta)^2$$

Least-Squares Geometry

$$e_i = y_i - \hat{y}_i$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n e_i^2$$

The full regression plots show residuals as vertical segments.
The toy plot isolates the squared-error pieces.

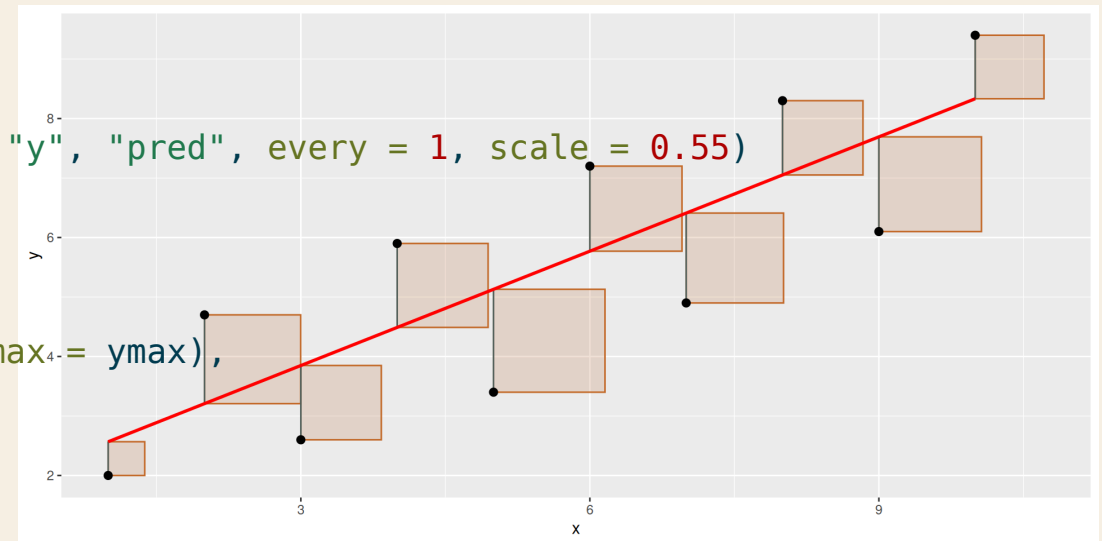
Toy Least-Squares Piece

```
toy_dat = data.frame(
  x = 1:10,
  y = c(2.0, 4.7, 2.6, 5.9, 3.4, 7.2, 4.9, 8.3, 6.1, 9.4)
)

toy_fit = lm(y ~ x, data = toy_dat)
toy_dat$pred = predict(toy_fit)

toy_squares = residual_square_data(toy_dat, "x", "y", "pred", every = 1, scale = 0.55)

ggplot(toy_dat, aes(x = x, y = y)) +
  geom_rect(
    data = toy_squares,
    aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
    inherit.aes = FALSE,
    fill = "#c46a2b",
    color = "#c46a2b",
    alpha = 0.18
  ) +
  geom_segment(aes(xend = x, yend = pred), color = "#155f7a", alpha = 0.65) +
  geom_point(size = 2.2) +
  geom_line(aes(y = pred), color = "red", linewidth = 1) +
  labs(x = "x", y = "y")
```



Feature / Design Matrix

A Feature or Design Matrix maps observations to individual feature criteria.

The most fundamental feature matrix is a column vector with all 1s, meaning every observation has the same expression for that feature.

In that case, the feature serves as a mapping to the overall mean of the observations.

Estimate The Mean With OLS

```
library(ggplot2)
library(knitr)
theme_set(theme_minimal(base_size = 16))

# load example data
data(airquality)

# our response vector is Wind
y = airquality$Wind

# create a design matrix for the mean
X = matrix(data = 1, nrow = length(y), ncol = 1)
colnames(X) = "Intercept"

str(X)
```

```
num [1:153, 1] 1 1 1 1 1 1 1 1 1 1 1 ...
- attr(*, "dimnames")=List of 2
 ..$ : NULL
 ..$ : chr "Intercept"
```

MSE Function

```
mse <- function(par, y, X) {  
  # the weights are stored in "par", the length of that vector depends  
  # on the number of columns in X (number of features)  
  
  # predicted value  
  pred = X %*% par  
  
  # mean squared error  
  mse = mean((y - pred)**2)  
  
  return(mse)  
}
```

Fit The Intercept

```
p <- rep(0, ncol(X))
names(p) <- colnames(X)

res <- optim(
  par = p,
  fn = mse,
  y = y,
  X = X
)

# our solution
res$par
```

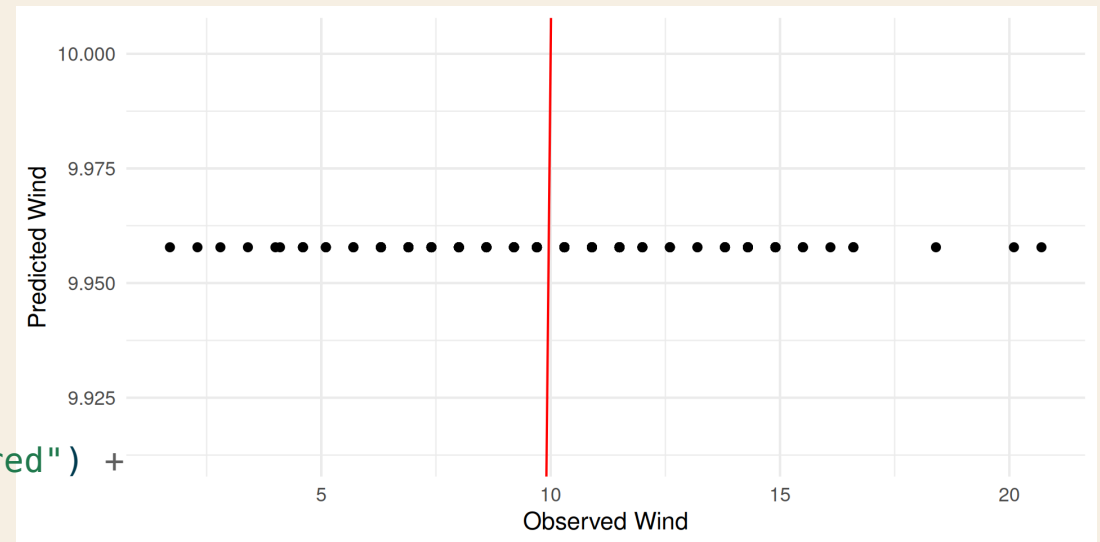
```
Intercept
9.957812
```

Predicted vs Observed

```
pred = X %*% res$par

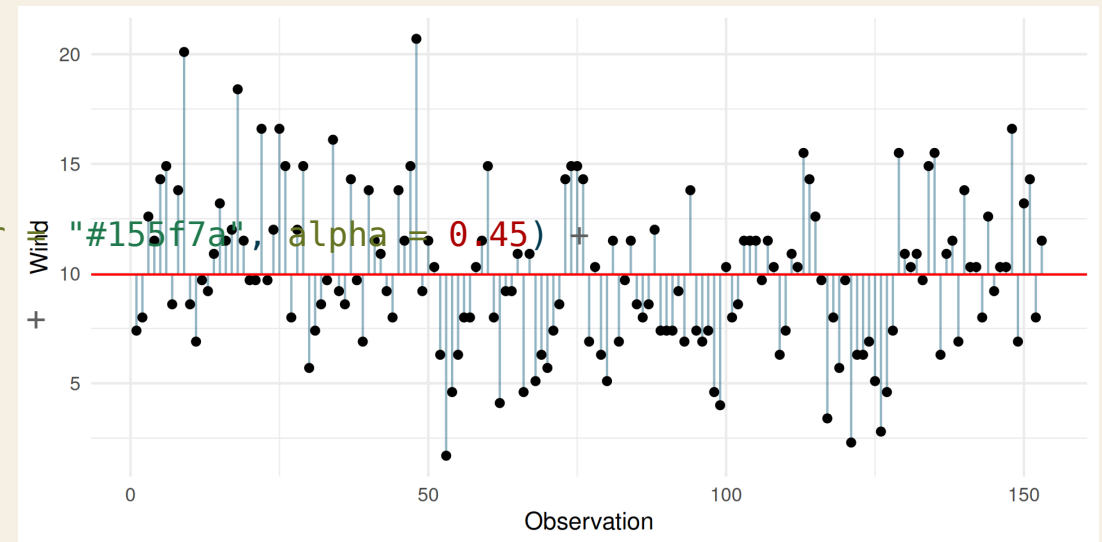
dat = data.frame(
  y = y,
  pred = as.numeric(pred),
  id = seq_along(y)
)

ggplot(dat, aes(x = y, y = pred)) +
  geom_point() +
  geom_abline(intercept = 0, slope = 1, color = "red") +
  labs(x = "Observed Wind", y = "Predicted Wind")
```



Raw Data And Intercept

```
ggplot(dat, aes(x = id, y = y)) +
  geom_segment(aes(xend = id, yend = pred), color = "#15577a", alpha = 0.45) +
  geom_point() +
  geom_hline(yintercept = res$par, color = "red") +
  labs(x = "Observation", y = "Wind")
```



Multiple Linear Regression

When there is more than one feature in the design matrix we call that Multiple Linear Regression.

The overall principle of the problem remains the same: We aim to minimize the mean squared error of our predictions.

Add Temperature

```
# our response vector is Wind
y = airquality$Wind

# now we also want another feature: Temp
temp = airquality$Temp

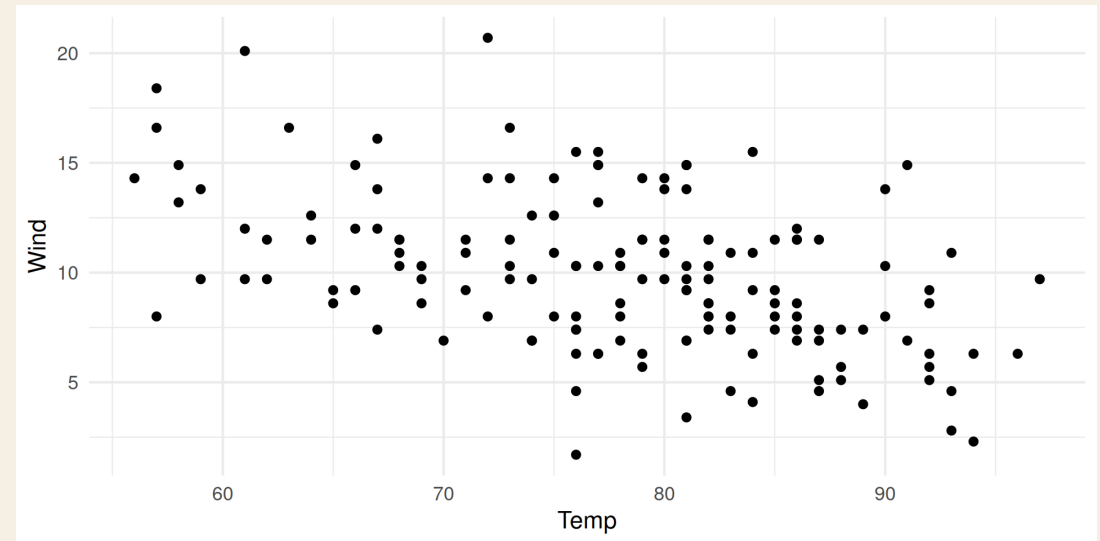
# create a design matrix for the mean
X = matrix(data = 1, nrow = length(y), ncol = 1)

# add temperature as a feature to our design matrix
X = cbind(
  X,
  temp
)

colnames(X) = c("Intercept", "Temp")
```

Wind vs Temp

```
ggplot(airquality, aes(x = Temp, y = Wind)) +  
  geom_point() +  
  labs(x = "Temp", y = "Wind")
```



Fit Multiple Regression

```
p <- rep(0, ncol(X))
names(p) <- colnames(X)

res <- optim(
  par = p,
  fn = mse,
  y = y,
  X = X
)

# our solution
res$par
```

Intercept	Temp
23.2314500	-0.1704349

Fitted Slope

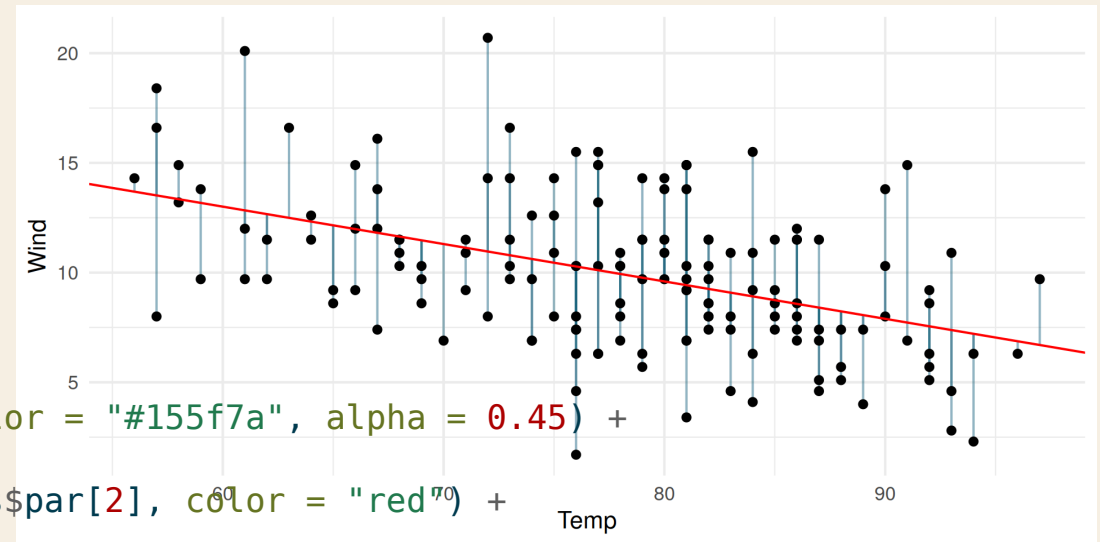
```

pred = X %*% res$par

dat = data.frame(
  Wind = y,
  Temp = airquality$Temp,
  pred = as.numeric(pred),
  id = 1:length(y)
)

ggplot(dat, aes(x = Temp, y = Wind)) +
  geom_segment(aes(xend = Temp, yend = pred), color = "#155f7a", alpha = 0.45) +
  geom_point() +
  geom_abline(intercept = res$par[1], slope = res$par[2], color = "red") +
  labs(x = "Temp", y = "Wind")

```



Ridge Regression

$$\min_{\beta} \left(\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right)$$

The ridge parameter λ controls how much we penalize large marker effects. We can visualize the solution space by evaluating the ridge criterion on a grid of λ values.

Closed-Form Ridge Solution

For a fixed λ , ridge regression has an analytical solution.

If $X = [\mathbf{1}, M]$ and the intercept is not penalized:

$$\hat{\beta}_\lambda = (X'X + \lambda P)^{-1} X'y, \quad P = \begin{bmatrix} 0 & 0 \\ 0 & I_p \end{bmatrix}.$$

With centered markers and centered response, as in the next code block:

$$\hat{u}_\lambda = \left(\frac{M'_c M_c}{n} + \lambda I_p \right)^{-1} \frac{M'_c y_c}{n}, \quad \hat{\beta}_0 = \bar{y}.$$

Lambda Solution Space

```
library(BGLR)
data(wheat)

subset = 1:100

y_ridge = wheat.Y[subset,2]
M_ridge = wheat.X[subset,1:300]

M_centered = scale(M_ridge, center = TRUE, scale = FALSE)
y_centered = y_ridge - mean(y_ridge)
n_obs = length(y_ridge)

fit_ridge_lambda <- function(lambda, y_centered, M_centered) {
  u = solve(
    crossprod(M_centered) / n_obs + diag(lambda, ncol(M_centered)),
    crossprod(M_centered, y_centered) / n_obs
  )

  pred = mean(y_ridge) + M_centered %*% u
  mse = mean((y_ridge - pred)**2)
  penalty = lambda * sum(u**2)

  data.frame(lambda = lambda, mse = mse, penalty = penalty,
             criteria = mse + penalty, sd_marker_effect = sd(as.numeric(u)))
}

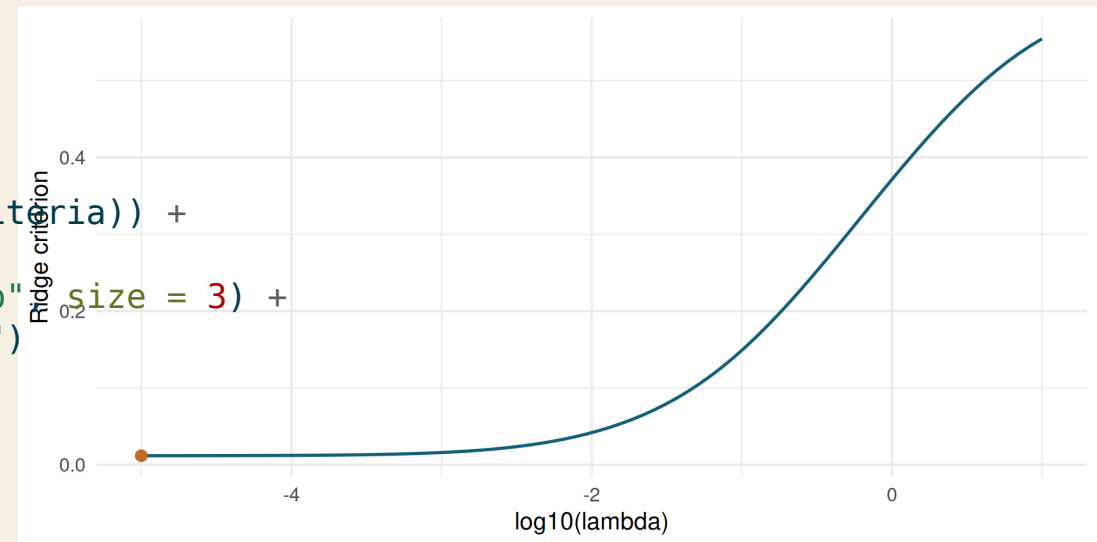
lambda_grid = 10^seq(-5, 1, length.out = 80)
ridge_path = do.call(rbind, lapply(lambda_grid, fit_ridge_lambda,
                                   y_centered = y_centered, M_centered = M_centered))

best_lambda = ridge_path[which.min(ridge_path$criteria), ]
best_lambda
```

	lambda	mse	penalty	criteria	sd_marker_effect
1	1e-05	0.01157643	4.447269e-05	0.0116209	0.1218662

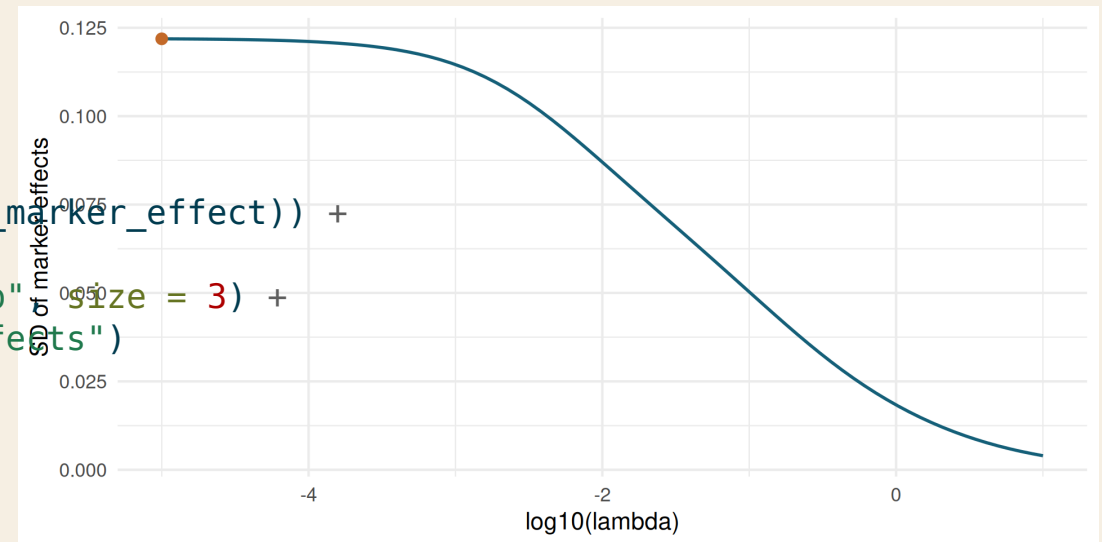
Ridge Criterion

```
ggplot(ridge_path, aes(x = log10(lambda), y = criteria)) +  
  geom_line(color = "#155f7a", linewidth = 1) +  
  geom_point(data = best_lambda, color = "#c46a2b", size = 3) +  
  labs(x = "log10(lambda)", y = "Ridge criterion")
```



Marker Shrinkage

```
ggplot(ridge_path, aes(x = log10(lambda), y = sd_marker_effect)) +  
  geom_line(color = "#155f7a", linewidth = 1) +  
  geom_point(data = best_lambda, color = "#c46a2b", size = 3) +  
  labs(x = "log10(lambda)", y = "SD of marker effects")
```



Ridge In The Original Lab

```
library(BGLR)
data(wheat)

subset = 1:100

y = wheat.Y[subset,2]
M = wheat.X
M = M[subset,1:300]

X = matrix(data = 1, nrow = length(y), ncol = 1)
colnames(X) = "intercept"
```

Ridge Objective

```
mse_ridge <- function(par, y, M, X) {  
  
  b = par[1:ncol(X)]  
  mu = X %*% b  
  
  u = par[(ncol(X) + 1):(length(par) - 1)]  
  pred = mu + M %*% u  
  
  mse = mean((y - pred)**2)  
  ridge_param = par[length(par)]  
  ridge = ridge_param * sum(u**2)  
  
  criteria = mse + ridge  
  
  if(ridge_param < 0) {  
    criteria = 9999999  
  }  
  
  return(criteria)  
}
```

Mixed Model Comparison

Now we compare the results from the optimization to an equivalent mixed model approach with a single random effect:

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + I\lambda \end{bmatrix} \begin{bmatrix} \beta \\ u \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

```
library(rrBLUP)
mod_rr = mixed.solve(y = y, X = X, Z = M)

ridge_mme = mod_rr$Ve / mod_rr$Vu
pred_mme = X %*% mod_rr$beta + M %*% mod_rr$u
```

Advanced Models

Mixed Model